

UCE823 :CLIMATE CHANGE AND ITS IMPACT ON WATER RESOURCES				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: To provide students with necessary knowledge to understand climate change and its impact on water resources with emphasis on hydrological cycle, adaption and mitigation strategies.				
Introduction: Basic definitions, greenhouse gases, global warming, weather and climate, climate variability and climate change, climate change scenarios over India Climate Change Modeling: Simple climate models, General Circulation models, concept of statistical downscaling Hydrological cycle and climate change: Tools and techniques for analysing changes in precipitation patterns and runoff. Effect of climate variability on:river flow variations leading to floods or droughts, glacier melt, sea level rise and urban flooding. Case studies on water crisis. Climate change adaptation: Role of indigenous traditional knowledge and resilience for future sustainability, different levels of adaption, adaption strategies, examples of climate change adaption in agriculture, urban planning, disaster preparedness including nature based solutions. Policy Framework for Climate Change Adaptation: Climate change policies, SDGs Approach, International Climate Change Agreements.				
Course Learning Objectives (CLO) Upon completion of this course, the students will be able to: 1. Define key terms associated with climate science, climate impacts, climate change policies 2. Download hydro-meterological data for climate change modelling 3. Analyse the impact of climate change on hydrological cycle and explain the effect of climate variability on water resources 4. Learn different adaption strategies and ways to address anticipated climate risks through policies and planning.				
Text Books 1. Brewster, E. N. <i>Climate Change Adaptation: Steps for a Vulnerable Planet</i>, New York, Nova Science, 2010 2. Rao Y.S.,Zhang, C.T., Ojha, C.S.P., Gurjar, B.R., Tyagi, R.D., and Kao, C.M. <i>Climate change modelling, mitigation and adaptation</i>, ASCE, 2013 3. Kevin E Trenberth. <i>Climate System Modeling</i>, Cambridge UniversityPress, 2010.				
Reference Books 1. Joseph, J.R. <i>Climate Change: What Everyone needs to know</i>, Oxford University Press, 2014 2. Kendal M. and Ann, H.S. <i>The climate modelling primer</i>, Wiley-Blackwell, 2014				

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	30
2.	EST	45
3.	Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations)	25